8 **(a)** In an α-particle scattering experiment, an α-particle is travelling in a vacuum towards the centre of a gold nucleus, as illustrated in Fig. 8.1.

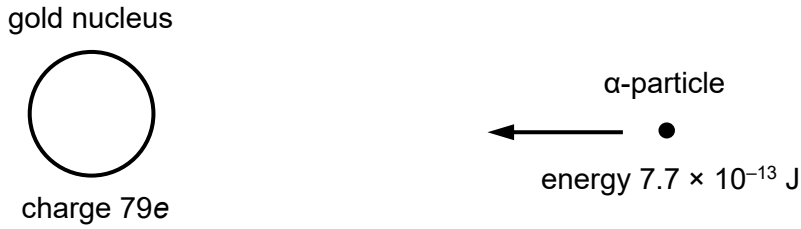

Fig. 8.1

The gold nucleus has a charge 79*e*. At a large distance from the gold nucleus, the α-particle has energy 7.7 × 10^{-13} J.

(i) The α-particle does not collide with the gold nucleus.

Show that the radius of the gold nucleus must be less than 4.7 × 10^{-14} m. [2]

(ii) The results of the α-particle scattering experiment provide evidence for the structure of the atom.

result 1: The vast majority of α-particles pass straight through the metal foil or are deviated by small angles.

result 2: A very small minority of α-particles are scattered through angles greater than 90° and up to 180°.

State what may be inferred from

1. result 1,

.....

..... [1]

2. result 2.

.....

.....

.....

..... [2]

(b) One nuclear reaction that can take place in a nuclear reactor may be represented, in part, by the equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{95}_{42}Mo + ^{139}_{57}La + 2\,^{1}_{0}n + \ .....\ + \text{ energy}$$

Data for a nucleus and some particles are given in Fig. 8.2

nucleus or particle	mass / u
$^{139}_{57}La$	138.955
$^{1}_{0}n$	1.00863
$^{1}_{1}p$	1.00728
$^{0}_{-1}e$	0.000549

Fig. 8.2

(i) Complete the nuclear reaction shown above. [1]

(ii) Calculate the binding energy per nucleon, in MeV, of lanthanum-139 $\left(^{139}_{57}La\right)$.

binding energy per nucleon = ... MeV [3]

(iii) State and explain whether the binding energy per nucleon of uranium-235 $\left({}^{235}_{92}\text{U}\right)$ will be greater, equal to or less than your answer to **(b)(ii)**.

.....

.....

.....

..... [2]

(c) A radiation detector is placed close to a radioactive source as shown in Fig. 8.3.

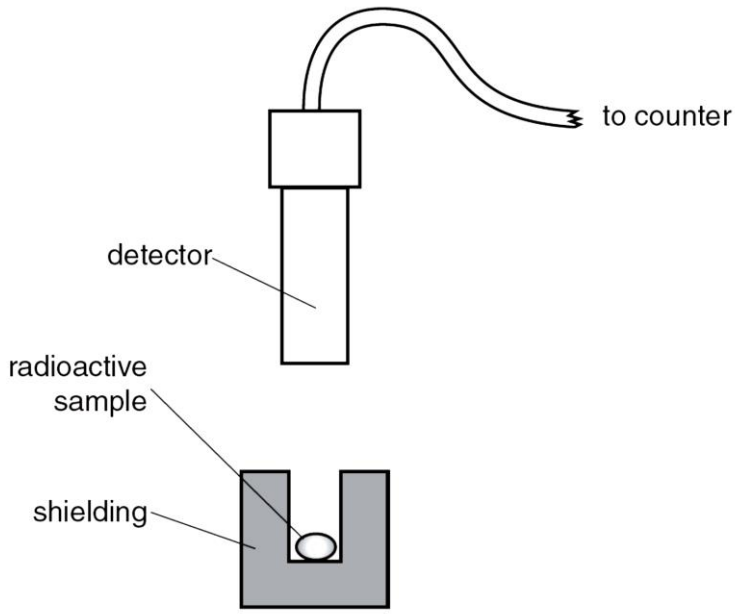

Fig. 8.3

The emissions from the radioactive source include both β-particles and γ-ray photons.
The β-particles emitted have energies up to a maximum of 0.61 MeV.

(i) Suggest two reasons why the activity of the source and the measured count rate may be different.

1. .....

..... [1]

2. .....

..... [1]

(ii) Explain why the emitted β-particles have a range of energies.

.....

..... [1]

(iii) The γ-ray photons emitted have specific energies. Suggest why this is so.

.....

..... [1]

(d) The variation with time *t* of the measured count rate in **(c)** is as shown in Fig. 8.4.

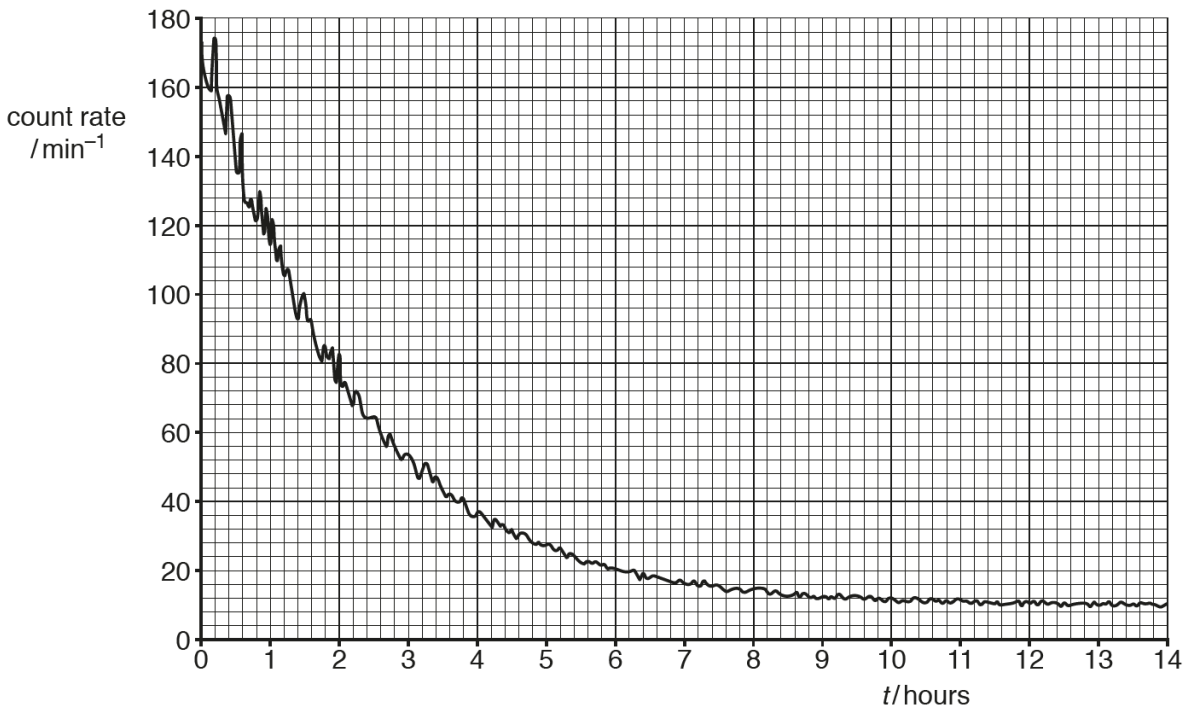

Fig. 8.4

(i) Use Fig. 8.4 to determine the half-life of the radioactive source.

half-life = ... hours [3]

(ii) The readings in Fig. 8.4 were obtained at room temperature.

A second sample of this radioactive source is heated to a temperature of 500 °C. The initial count rate at time t = 0 is the same as that in Fig. 8.4. The variation with time t of the measured count rate from the heated source is determined.

State and explain if there are any differences in

1. the half-life,

.....

..... [1]

2. the measured count rate for any specific time.

.....

..... [1]